



Designing Authentic Industry-Engaged Assessment for Professional Competence in Business Intelligence

Dr. Helga Ingimundardóttir <helgaingim@hi.is>

22nd International CDIO Conference: June 22nd 2026

INDUSTRIAL ENGINEERING

2026-04-30

helgaingim CDIO: BI redesign

Designing Authentic Industry-Engaged Assessment
for Professional Competence in Business Intelligence

Dr. Helga Ingimundardóttir (helgaingim@hi.is)

22nd International CDIO Conference: June 22nd 2026

INDUSTRIAL ENGINEERING

Motivation



Academia

Aims to prepare students for professional practice, but offers limited guidance on how to operationalize this in course design and assessment

Industry

Expects credible, reproducible results supported by clear reasoning and usable for decision-making

Underlying mismatch

What counts as “good work” differs between academic and professional contexts

2026-04-30

helgaingim CDIO: BI redesign

└ Background

└ Motivation

This work starts from a familiar gap.

Universities aim to prepare students for professional practice – but translating that into course design is not straightforward.

At the same time, industry expectations are quite concrete. Work needs to be credible, reproducible, and usable.

And these perspectives do not always align.

So the question is not whether we value the same things – but how those values are actually operationalized in courses.

Motivation	
	Academia
Aims to prepare students for professional practice, but offers limited guidance on how to operationalize this in course design and assessment	
	Industry
Expects credible, reproducible results supported by clear reasoning and usable for decision-making	
	Underlying mismatch
What counts as “good work” differs between academic and professional contexts	

What prompted the redesign



Observed in an industry-facing course

- Industry partners valued collaboration – but would not use the results
- Students reported a very heavy and uneven workload

Core question

How do we design courses so that student work is **authentic in practice**, while still supporting **meaningful learning**?

2026-04-30

helgaingim CDIO: BI redesign

└ Background

└ What prompted the redesign

This is not just a theoretical issue.

In an earlier version of the course, we worked with an industry partner and asked how useful the student results were.

They appreciated the collaboration – but they would not actually use the results.

At the same time, students experienced the course as quite heavy and time-consuming.

So the course was not fully working for either side.

That leads to the core question:

how do we design assessment so that the work is **authentic in practice**, while still supporting **meaningful learning**?

What prompted the redesign



Observed in an industry-facing course

- Industry partners valued collaboration – but would not use the results
- Students reported a very heavy and uneven workload

Core question

How do we design courses so that student work is **authentic in practice**, while still supporting **meaningful learning**?

Context: Business Intelligence course



Overview

- ▶ 6 ECTS course
 - ▶ advanced undergraduate
 - ▶ early graduate
- ▶ 10–15 students, two teams

What is BI?

Data-driven decision support: combining data analysis, systems, and communication to inform organizational decisions.

Goal

Strengthen industrial relevance and professional competence

helgaingim CDIO: BI redesign

└ Background

└ Context: Business Intelligence course

2026-04-30

The context is a Business Intelligence course.

What matters here is that BI is not just about analysis – it is about **decision support**.

So students are not only expected to produce correct analyses, but also to communicate assumptions, limitations, and implications.

And importantly, this is done in collaboration with an **industry partner**, so the work is evaluated in a realistic context.

Context: Business Intelligence course



Overview

- ▶ 6 ECTS course
- ▶ advanced undergraduate
- ▶ early graduate
- ▶ 10–15 students, two teams

What is BI?

Data-driven decision support: combining data analysis, systems, and communication to inform organizational decisions.

Goal

Strengthen industrial relevance and professional competence

Industry-engaged project



Sustained industrial context

A single dataset and domain used across the entire semester

Industry partner role

- ▶ problem framing
- ▶ milestone discussions
- ▶ final evaluation

Dual context

Academic and industry perspectives coexist

helgaingim CDIO: BI redesign

└ Background

└ Industry-engaged project

2026-04-30

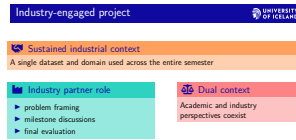
An initial, relatively simple change addressed both concerns.

We moved to a **single, persistent industry case** across the semester.

This reduces workload by removing repeated onboarding and allows students to build a deeper and more coherent understanding of the data and domain.

So instead of producing isolated analyses, they develop work that evolves over time within a realistic setting.

This creates the conditions for more meaningful and usable results.



Design focus: LOUIS-guided

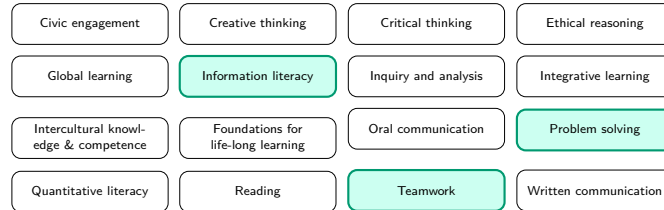


Design focus

- ▶ limited set of competences (LOUIS)
- ▶ sustained industry context
- ▶ iterative development toward capstone

LOUIS framework

Learning Outcomes in University for Impact on Society
Focus on a small set of competences (max 3)



helgaingim CDIO: BI redesign
└ Design approach and structure

└ Design focus: LOUIS-guided

For this redesign, I attended a workshop on the LOUIS framework – **Learning Outcomes in University for Impact on Society**.

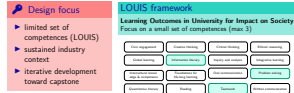
LOUIS is part of the Aurora University Network, and is intended to strengthen broader academic and personal competences alongside subject knowledge.

A key takeaway from that workshop was to avoid spreading too thin – instead, focus on a **small number of competences** and make expectations explicit.

So this redesign is directly informed by that work.

LOUIS helped translate high-level competence goals into concrete learning outcomes that students can understand and act on.

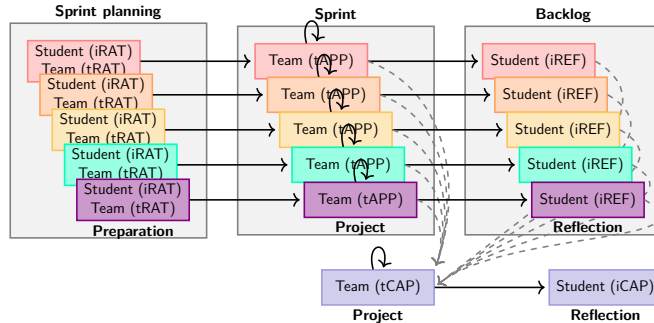
Design focus: LOUIS-guided



Pedagogical structure

Week 1–2
Data Ethics
Week 3–4
Large Language Models
Week 5–6
Supervised Learning
Week 7–8
Clustering
Week 9–10
Process Mining

Week 11–14
Capstone Project



+ Team-Based Learning (TBL):

Flipped classroom

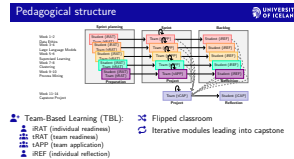
Iterative modules leading into capstone

- iRAT (individual readiness)
- tRAT (team readiness)
- tAPP (team application)
- iREF (individual reflection)

helgaingim CDIO: BI redesign
Design approach and structure

Pedagogical structure

2026-04-30



The course combines **TBL** with **iterative** project work via **PBL**

Students prepare individually before class, and class time is used for discussion and guided project work. (flipped classroom)

If we translate this into an Agile perspective:

- The preparation and initial team phase function like **sprint planning**.
- The subsequent sessions correspond to the **sprint itself**, where teams develop and refine their work.
- And the individual reflection acts as a **retrospective**, capturing what should be improved next.
- The capstone then brings this together. Rather than starting a new cycle, students revisit and refine previous work – which is closer to what in Agile would be called **iteration on an evolving product**, not isolated sprints.
- The final course reflection then acts as a broader retrospective.

Core mechanism: assessment as development



☰ One rubric, many iterations

Same assessment rubric reused across all modules and the capstone

Standards fixed; performance expectations scaled during learning

☑ Assessment criteria

- ▶ analytical soundness
- ▶ transparency
- ▶ reproducibility
- ▶ stakeholder relevance

▶▶ Developmental shift

Feedback is **carried forward**
— *not* reset between tasks

helgaingim CDIO: BI redesign

└ Expectations and feedback

└ Core mechanism: assessment as development

2026-04-30

A key mechanism is how expectations are made explicit.

From the beginning, students are shown the **final rubric** – so they know what high-quality work looks like.

However, during the learning phase, assessment is **scaled**.

So instead of grading against the final standard, work is normalized to what counts as “**very good**”.

This signals clearly: this is what is expected for the final project.

It also helps students prioritize their effort – they can identify what needs to be improved and revisit earlier work.

So the rubric is not just used for grading – it becomes a tool for **guiding development over time**.

This is important because it shifts effort from producing isolated outputs to improving work toward something that could actually be used in practice.

Core mechanism: assessment as development

☰ One rubric, many iterations

Same assessment rubric reused across all modules and the capstone
Standards fixed; performance expectations scaled during learning

☑ Assessment criteria

- ▶ analytical soundness
- ▶ transparency
- ▶ reproducibility
- ▶ stakeholder relevance

▶▶ Developmental shift

Feedback is **carried forward**
— *not* reset between tasks

Accumulated feedback



Pull Requests as assessment architecture

Changes are proposed, reviewed, discussed, and integrated – or explicitly deferred

Structured accumulation

- ▶ same rubric reused
- ▶ prior feedback revisited
- ▶ unresolved issues explicitly carried forward

Enabled by tooling

-  version control
-  persistent work traces

2026-04-30

helgaingim CDIO: BI redesign
└─ Expectations and feedback

└─ Accumulated feedback

In practice, this is implemented through **pull requests**.

Each piece of work is proposed, reviewed, discussed, and then either accepted or deferred. So instead of submitting assignments and moving on, the work remains **open and revisitable**. If an issue is raised and not resolved, it stays visible.

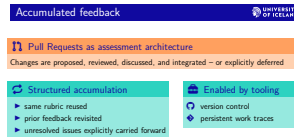
That means later work is evaluated in light of earlier feedback.

Students cannot simply move on – they have to **revisit and improve**.

Because the same rubric is reused, I can also directly see what has changed – and whether earlier issues have been addressed.

So assessment becomes more focused, and grading is faster because I am not re-evaluating everything from scratch.

This creates continuity across the course, and is much closer to how work functions in professional settings.



Evolving feedback



Instructor

- ▶ method
- ▶ implementation

Industry partner

- ▶ domain relevance
- ▶ decision context

Student

- ▶ knowledge transfer through PRs
- ▶ defending decisions
- ▶ critiquing peer work

Progression over time

1. Align within teams
2. Cross-team review
3. Industry validation

Audience shift

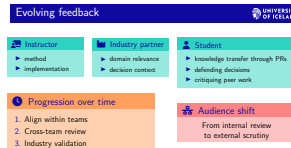
From internal review
to external scrutiny

helgaingim CDIO: BI redesign
Expectations and feedback

2026-04-30

Evolving feedback

1. **Who** gives feedback:
 - The instructor focuses on method and implementation.
 - The industry partner focuses on relevance and decision context.
 - And importantly, students themselves are a key part of the process. Through PRs, they engage in **knowledge transfer** – by defending their own work and critiquing others.
2. **When** does the feedback evolve:
 - During the modules, the focus is on aligning within teams.
 - In the capstone, this expands to cross-team review, bringing in a fresh perspective.
 - And finally, the work is exposed to the industry partner.
3. **What this means** is the audience widens:
 - Feedback moves from internal to external and more critical scrutiny.
 - And students are expected to respond to all of these – not just receive them.



Credibility and trust



Industry threshold

Credibility is fragile: early issues can undermine trust

What is evaluated

- ▶ reproducibility
- ▶ transparency
- ▶ clear assumptions

Shift in communication

Build trust first
– then invite inspection

helgaingim CDIO: BI redesign

Findings

Credibility and trust

2026-04-30

As feedback moves from internal to external, the consequences change.

In an earlier version of the course, students presented their work in a standard academic way. During one presentation, the industry partner identified an issue early in the analysis. Even though the rest of the work was reasonable, that early issue reduced trust in the entire result.

So they did not engage further.

This is what we mean by credibility being fragile.

So we made a deliberate shift in how students communicate.

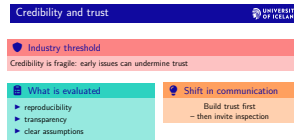
The goal is to **build trust first**, so the audience is willing to engage more deeply.

That means clearly presenting results, assumptions, and what can be done with the analysis.

The detailed reasoning is still there – but supported through the report and the codebase.

Because the full history is available, the work becomes **traceable and reproducible**.

So credibility is not just claimed – it can be inspected.



Balancing team and individual assessment



Team-level assessment

- ▶ analytical quality
- ▶ coherence
- ▶ relevance

Individual-level assessment

- ▶ visible artifacts
- ▶ documented contributions
- ▶ structured reflection

Design goal

Accountability without fragmenting teamwork

Designed into the course architecture—not added after the fact

helgaingim CDIO: BI redesign

Findings

Balancing team and individual assessment

2026-04-30

Balancing team and individual assessment



Team-level assessment

- ▶ analytical quality
- ▶ coherence
- ▶ relevance

Individual-level assessment

- ▶ visible artifacts
- ▶ documented contributions
- ▶ structured reflection

Design goal

Accountability without fragmenting teamwork
Designed into the course architecture—not added after the fact

Another design challenge is balancing teamwork with individual accountability. The goal is not to split the work into isolated individual tasks, but to make individual contributions **visible within the team context**. This is enabled through the workflow we've introduced. Because all work is proposed, reviewed, and discussed, you can see who contributed what, how they engaged with feedback, and how their work evolved over time. So accountability comes from **traceable contributions**, not from separating the work. This allows us to maintain strong collaboration while still assessing individual engagement.

Students & graduates

- ▶ high engagement without attendance requirements
- ▶ transfer of reproducible workflows
- ▶ stronger documentation practices

Academic and industry perspectives

-  more ambitious, student-driven projects
-  improved coherence and credibility

There is no formal attendance requirement in this course. However, compared to other courses I teach with the same student group, attendance and participation are noticeably higher.

In more conventional formats – lectures, “regular” industry-facing final project – engagement tends to be lower.

Here, students are more actively involved throughout the process.

We also observe transfer. Students adopt reproducible workflows and stronger documentation practices, and continue to use them beyond the course.

After graduation, students often reflect on this format as one of the most impactful parts of their studies – not just in terms of content, but in how they learned to work.

From the industry perspective, the outputs are more coherent and credible, and importantly, they can see how the results could be implemented in practice.

So while this is not controlled evidence, the signal is consistent across students, graduates, and partners.

Students & graduates	Academic and industry perspectives
▶ high engagement without attendance requirements ▶ transfer of reproducible workflows ▶ stronger documentation practices	▶ more ambitious, student-driven projects ▶ improved coherence and credibility

What students must learn to handle



⚡ What becomes visible

The design makes certain challenges explicit

- ▶ giving and receiving critical feedback
- ▶ making work visible and reviewable
- ▶ managing workload in a structured workflow

helgaingim CDIO: BI redesign

Findings

What students must learn to handle

What becomes clear with this design is not just improved outcomes – but also the challenges students need to learn to handle.

1. **Giving and receiving critical feedback:** Students are often hesitant to critique peers, but this becomes a central part of the workflow.
2. **Making work visible:** Everything is open to review – which creates accountability, but also requires students to articulate and justify their decisions.
3. **Managing workload:** The structure introduces overhead, and students feel that early on.

But that structure is also what enables iteration, feedback, and improvement.

So the point is not to remove these challenges.

They are a direct consequence of making the work visible – and part of what students need to learn to navigate.

What students must learn to handle



⚡ What becomes visible

The design makes certain challenges explicit

- ▶ giving and receiving critical feedback
- ▶ making work visible and reviewable
- ▶ managing workload in a structured workflow

What this design does



Transferable principles

For CDIO-aligned, industry-engaged courses

- ▶ sustained industry context (no repeated onboarding)
- ▶ assessment drives iteration (not just evaluation)
- ▶ contributions are visible and reviewable
- ▶ feedback accumulates across the course
- ▶ students adapt to different audiences
- ▶ tooling supports – but does not define – the design

2026-04-30

helgaingim CDIO: BI redesign

└ Summary

└ What this design does

This can be summarized in a few key design choices.

1. First, a sustained industry context – so students build depth instead of repeatedly onboarding.
2. Second, assessment is used to drive iteration, not just to evaluate final outputs.
3. Third, contributions are made visible and reviewable, which supports accountability without fragmenting teamwork.
4. Fourth, feedback accumulates across the course, so students revisit and improve earlier work.
5. And finally, students learn to adapt their work to different audiences – academic and industry.

The tooling supports this, but the key contribution is the alignment between learning and professional practice.

What this design does



Transferable principles

For CDIO-aligned, industry-engaged courses

- ▶ sustained industry context (no repeated onboarding)
- ▶ assessment drives iteration (not just evaluation)
- ▶ contributions are visible and reviewable
- ▶ feedback accumulates across the course
- ▶ students adapt to different audiences
- ▶ tooling supports – but does not define – the design

Current limitations

- ▶ depends on student readiness
- ▶ scalability considerations
- ▶ domain-specific transferability

Design refinements

- ▶ earlier preparation for professional tooling
- ▶ improved stakeholder interaction
- ▶ structured team-building to accelerate trust in large groups
- ▶ scaling across courses and partners

Limitations and future directions

This is a design-oriented case – not a controlled study.

It depends on student readiness, and there is some initial overhead.

Early in the semester, students often experience the workload as heavy. But later they consistently describe this as one of the most valuable parts of their studies.

So there is a clear shift from short-term effort to long-term value.

From a teaching perspective, this creates a very **dynamic environment** – but it also requires hands-on interaction. So scaling needs to be considered carefully. At the current size, it is manageable.

In terms of transferability, this fits naturally within BI – but the underlying design can extend to other courses. But it requires both students and instructors to commit to the structure.

That is why we are now introducing tooling and practices earlier in the curriculum.

And one ongoing refinement is strengthening the initial phase – giving students more time to build trust, understand the data, and align as a team – before introducing more advanced methods.

Current limitations

- ▶ depends on student readiness
- ▶ scalability considerations
- ▶ domain-specific transferability

Design refinements

- ▶ earlier preparation for professional tooling
- ▶ improved stakeholder interaction
- ▶ structured team-building to accelerate trust in large groups
- ▶ scaling across courses and partners

Questions?



Contact details:

- ✉ helgaingim@hi.is
- ☎ +354 525-4630
- 🏛 VR-II, University of Iceland
- 🌐 helgaingimundardottir
- 🐙 @tungufoss

This brings us back to the original problem.

The issue was not student ability – it was that the work was not credible or usable in context.

What this case shows is that professional competence does not emerge by itself.

It requires making expectations explicit – especially around credibility, trust, and use.

Assessment and feedback are the mechanisms that make this visible.

When feedback accumulates, and when students must respond to different audiences, they begin to align their work with real practice.

So the contribution is not a specific tool or method.

It is showing how assessment and feedback can be used to align learning with professional expectations.

